

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
CONCEPT RECAPITULATION TEST – III
PAPER –1
TEST DATE: 31-07-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 – 04, 20 – 23, 39 – 42): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
Section – A (05 –10, 24 – 29, 43 – 48): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

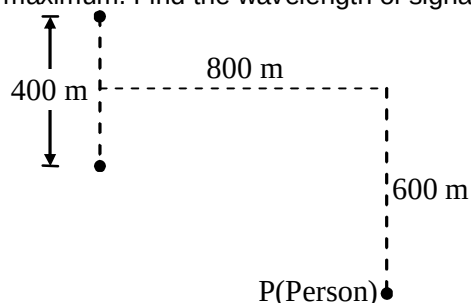
Physics

PART – I

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

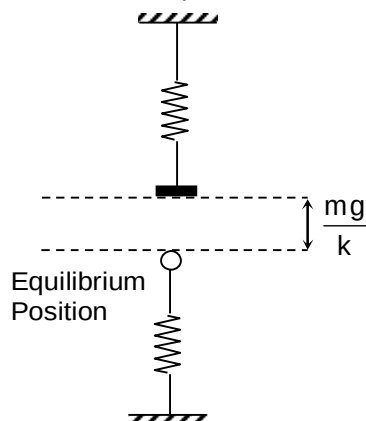
1. Figure shows radio antennas kept at a distance 400m. The antennas broadcast the same signals at the same wavelength. A person, having a radio, is standing at the position of third order maximum. Find the wavelength of signals.



- (A) 60 m
(B) 79 m
(C) 89 m
(D) 100 m
2. A body is projected vertically upwards from the surface of earth with a velocity sufficient to carry it to infinity. Calculate the time taken by it to reach height h (Given radius of earth = R).
- (A) $\frac{1}{3} \sqrt{\frac{2R}{g}} \left[\left(1 + \frac{h}{R} \right)^{3/2} - 1 \right]$
(B) $\frac{1}{2} \sqrt{\frac{2R}{g}} \left[\left(1 + \frac{h}{R} \right)^{3/2} - 1 \right]$
(C) $\frac{1}{3} \sqrt{\frac{R}{g}} \left[\left(1 + \frac{h}{R} \right)^{3/2} - 1 \right]$
(D) $\sqrt{\frac{R}{g}} \left[\left(1 + \frac{h}{R} \right)^{3/2} - 1 \right]$
3. Find the decay life time of ^{55}Co radio nuclide if its activity is known to decrease 4% per hour. The decay product is non-radioactive ($\ln 0.96 = -0.041$).
- (A) $\lambda \approx 2.1 \times 10^{-5} \text{ s}^{-1}$
(B) $\lambda \approx 2.3 \times 10^{-5} \text{ s}^{-1}$
(C) $\lambda \approx 3.2 \times 10^{-5} \text{ s}^{-1}$
(D) $\lambda \approx 1.1 \times 10^{-5} \text{ s}^{-1}$

4. Figure shows a small block of mass m attached to a spring of force constant k and an identical spring hangs from ceiling. Initially lower spring is in compressed state with compression equal to $\frac{3mg}{k}$ from natural length of spring. When block is released it strikes upper spring and sticks to it.

What is the amplitude of oscillation (in cm) after sticking? (Given: $mg = 10 \text{ N}$, $k = 100\sqrt{7} \text{ N/m}$)

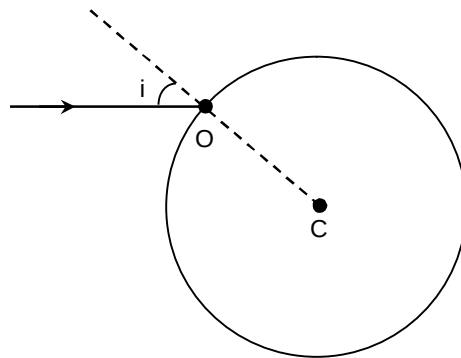


- (A) 5 cm
(B) 4 cm
(C) 6 cm
(D) 7 cm

Section – A (Maximum Marks: 24)

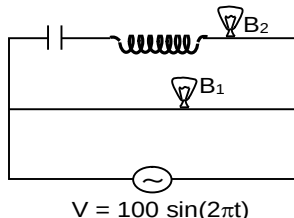
This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. A spherical transparent medium of refractive index n is placed in air. A thin parallel beam of light of monochromatic wavelength λ is incident at a point O of the sphere at an angle of incident i . Then which of the following statements will be true for this optic system?

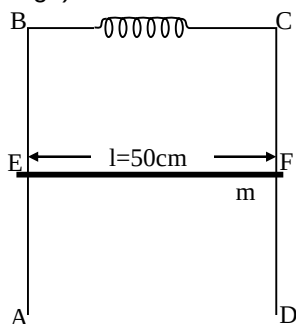


- (A) For the ray to come out of the sphere after refraction through it, the incident angle i should be less than the critical angle C for the medium of the sphere.
(B) The angle of emergence θ (made with the normal) for the ray will be equal to the incident angle i for all values of i .
(C) If the angle $i = 90^\circ$ (grazing incidence) the emergent ray will come out grazing the surface.
(D) If the angle of incident i is such that the angle of refraction at O is equal to the critical angle C for the medium of sphere, the emergent ray will be parallel to the incident ray.

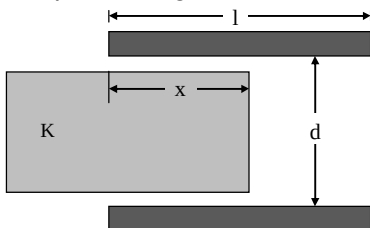
6. Two similar bulbs of rating $(100\text{W}, 50\sqrt{2}\text{ V})$ are connected in circuit as shown. When frequency of source is varied, brightness of B_2 changes. It is observed that, at 50 Hz frequency both the bulbs have same brightness. Select correct alternative(s).



- (A) Current in both the bulbs will be in same phase when both have same brightness.
 (B) RMS current through source will be 2 A when both the bulbs have same brightness.
 (C) RMS current through source will be $2\sqrt{2}$ A when both the bulbs have same brightness.
 (D) B_1 will be more brighter for all frequencies except 50 Hz.
7. A conducting frame ABCD is kept in a vertical plane. A conducting rod EF of mass 1 kg and length 50cm can slide smoothly on it remaining horizontal always. The resistance of the frame is negligible and inductance is constant having value 1H. The rod left from rest and allowed to fall under gravity with no current in the inductor. A magnetic field of constant magnitude 2T is present throughout the loop pointing inwards. Then (Assume length AB and CD of frame is sufficiently large):



- (A) Position of the rod as a function of time assuming initial position of the rod to be $x = 0$ and vertically downward as the positive x-axis is $x = 10[1 - \cos t]$.
 (B) Maximum current in the circuit is 20A.
 (C) Maximum velocity of the rod is 20 m/s.
 (D) After some time velocity of rod becomes constant.
8. Two square plates of sides ℓ are placed parallel to each other with separation d as suggested in figure. You may assume d is much less than ℓ . The plates carry uniformly distributed static charges $+Q_0$ and $-Q_0$. A block of metal has width ℓ , length ℓ , and thickness slightly less than d . It is inserted at distance x into the space between the plates. The charges on the plates remain uniformly distributed as the block slides in. In a static situation, a metal prevents an electrical field from penetrating inside it. The metal can be thought of as a perfect dielectric, with $K \rightarrow \infty$.

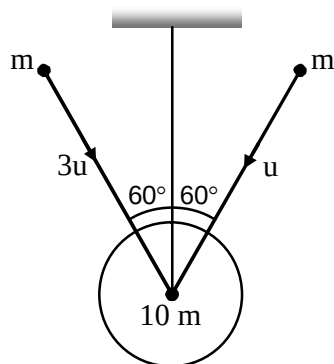


- (A) The stored energy as a function of x is given by $\frac{Q_0^2 d(\ell - x)}{(2\ell^3 \epsilon_0)}$.
- (B) The magnitude of the force that acts on the metallic block is given by $\frac{Q_0^2 d}{(\ell^3 \epsilon_0)}$ to the right.
- (C) The area of the advancing front face of the block is essentially equal to ℓd , the stress (force per unit area) on it is $\frac{Q_0^2}{(2\ell^4 \epsilon_0)}$.
- (D) The energy density in the electric field between the charged plates in terms of Q_0, ℓ, d , and ϵ_0 is $\frac{Q_0^2}{(2\ell^4 \epsilon_0)}$.

9. The vibration of a string of length 60 cm fixed at both ends are represented by the equation $y = 4 \sin \frac{\pi x}{15} \cos(96\pi t)$, where x and y are in cm and t in second. Then:

- (A) maximum displacement of a particle at a point at $x = 5$ cm is 4 cm
- (B) third node is formed at 30 cm from one end of the string
- (C) the velocity of a particle at $x = 7.5$ cm and $t = 0.25$ s is zero
- (D) equations of the component waves whose superposition gives the above equation are $y_1 = 2 \sin\left(\frac{\pi x}{15} - 96\pi t\right)$ and $y_2 = 2 \sin\left(\frac{\pi x}{15} + 96\pi t\right)$

10. A bob of mass $10m$ is suspended through an inextensible string of length l . When the bob is at rest in equilibrium position, two particles each of mass m strike it as shown in figure. The particles stick after collision. Choose the correct statement from the following:

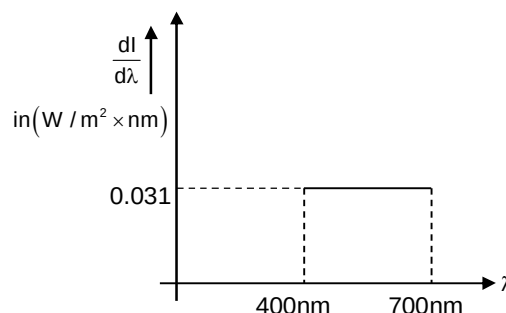


- (A) Impulse in the string due to tension is $2mu$
- (B) Velocity of the system just after collision is $v = \frac{u\sqrt{3}}{12}$
- (C) Loss of mechanical energy of the system is $\frac{137}{24}mu^2$
- (D) Loss of mechanical energy of the system is $\frac{137}{48}mu^2$

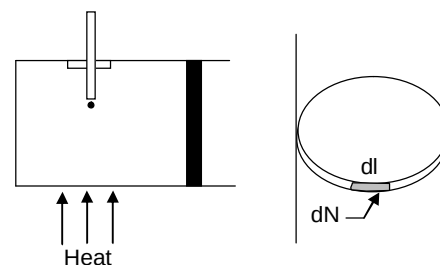
Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

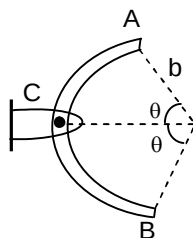
11. Scientists have made a light source whose spectral emissive power $\frac{dI}{d\lambda}$ is constant over visible range. Here I is intensity and λ is wavelength. In other words, $\frac{dI}{d\lambda}$ graph is shown below. This beam of area 1 m^2 is incident on emitter plate of photoelectric tube. The collector plate is sufficiently positive so that tube is in saturation mode. Assume that each capable photon liberates 1 electron. If work function is 2 eV, what is the current (in A) in the tube? Round off to nearest integer.



12. Figure shows a cylindrical tube of radius 5 cm and length 20 cm. It is closed by tight fitting cork. The friction coefficient between the cork and the tube is 0.20. The tube contains an ideal gas at a pressure of 1 atm and temperature 300K. The tube is slowly heated and it is found that the cork pops out when the temperature reaches 600 K. Let dN denote the magnitude of normal contact force exerted by small length dl of the cork along the periphery. Assuming that the temperature of gas is uniform at any instant, the value of $\frac{dN}{dl} = 25 \times \alpha \times 10^\beta$. Find the value of $\alpha + \beta$.



13. A uniform rod AB bent in the shape of an arc of circle is free to rotate in a vertical plane about a fixed horizontal axis passing through the mid-point of the wire. The situation is shown in the figure. The angular acceleration of the rod immediately after it is released from rest from the position shown is $\alpha = \frac{pg}{4b}$. Calculate p .

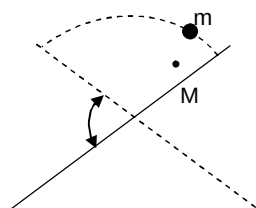


Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15
Question Stem

A homogeneous bar of mass M rotates uniformly in alternating directions to and fro around the horizontal axis going through its center, while a ball of mass m bounces between the ends of the bar in vertical plane as shown in the figure. The angular velocity of the bar is $\pm \frac{2\pi}{3}$ rad sec⁻¹ and the time period of the periodic motion of the ball is 2 second. (The mechanical loss and the duration of the collisions are negligible.)



14. What is the velocity (in m/s) of the ball at the highest point of its orbit?
15. How long (in meter) is the bar?

Question Stem for Question Nos. 16 and 17
Question Stem

One of the most fierce battles in the history of humanity was the battle between Arjuna and Karna at Kurukshetra. One night, Karna conceived of a plan to defeat and kill Arjuna in the war. He prepared an arrow with a charge of +1 C deposited on its tip. From his study of circuits at school days, he estimated that if this arrow hits the crown of Arjuna, this charge will be conducted to the ground in almost 0.001 sec. The arrow has a mass of 100 gm. Arjuna would be shocked to death. Not only that, the heat that would be produced would burn his body to ashes.

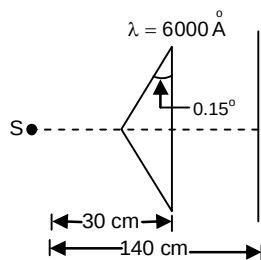
But Karna was not aware that Arjuna was being protected personally by Supreme Lord Sri Krishna. Lord Krishna knew of this plan and he devised a simple plan to counter it. As Karna would fire his arrow, the Lord would build a uniform magnetic field directed vertically so that the arrow would be deviated and would not touch Arjuna. In fact, the magnetic field could be such that the arrow would come back and hit Karna.

16. Assume that Arjuna is standing 10 m east from Karna. The arrow is fired with a horizontal velocity of 50 m/s. What is the magnetic field so that the arrow deviates by 90° by the time it travels a distance of 10 m in the east direction? Neglect gravity for this question only.
17. Actually gravity cannot be neglected and the arrow would follow a helical path, eventually falling to the ground. If Karna has fired the arrow horizontally from a height of 1.8m, what is the angle in radians that the arrow turns before it hits the ground? (Velocity of projection and magnetic fields remains same)

Question Stem for Question Nos. 18 and 19

Question Stem

Refracting angles of biprism are both 0.15° . Distance of biprism from source is 30 cm. Distance between source and screen is 140 cm. The Fresnel's biprism arrangement shown in the figure uses a source of light of wavelength $6000 \text{ \AA}$. When a converging lens moves between the biprism and the screen two images of S are focused on the screen for two positions of the lens. The separations between the two images for the two positions are found to be 0.9 mm and 1.6 mm.



18. The refractive index of prism is ?
19. The fringe width of the pattern on screen (after the lens has been removed) is:

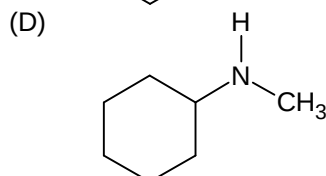
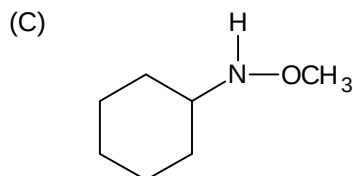
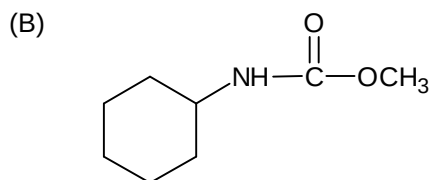
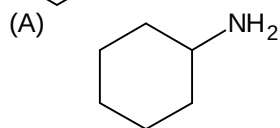
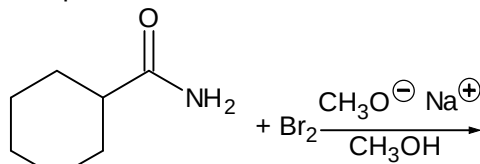
Chemistry

PART – II

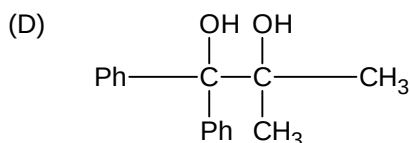
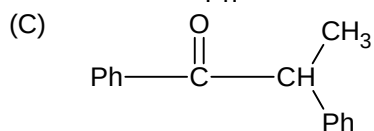
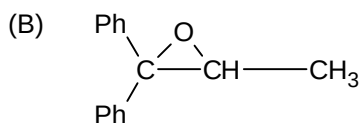
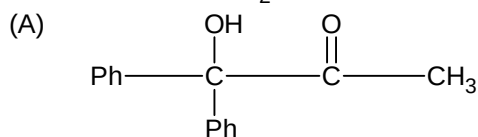
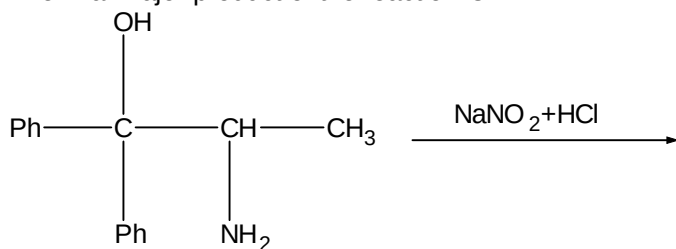
Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

20. The product formed in the reaction is :



21. The final major product of the reaction is :



22. $\text{Pt} \mid \text{Cl}_2 (P_1 \text{ atm}) \mid \text{HCl} (0.1 \text{ M}) \mid \text{Cl}_2 (P_2 \text{ atm}) \mid \text{Pt}$, cell reaction will be spontaneous if:

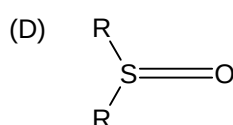
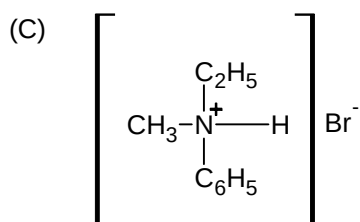
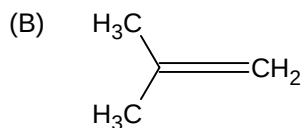
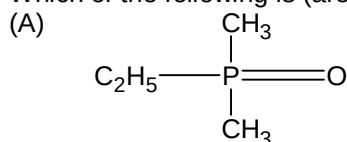
- (A) $P_1 = P_2$
 (B) $P_1 > P_2$
 (C) $P_2 > P_1$
 (D) $P_1 = P_2 = 1 \text{ atm}$

23. A liquid system consists of two immiscible liquids water and n- butyl chloride distills at one atmosphere pressure. If vapour pressure of water at that temperature is 570 mm of Hg, then the ratio of masses of water and n- butyl chloride distilled is :
- (A) 3 : 1
 (B) 0.58 : 1
 (C) 1 : 1
 (D) 1 : 3

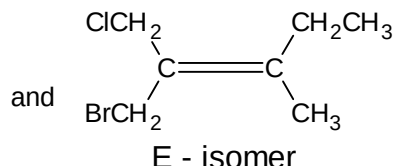
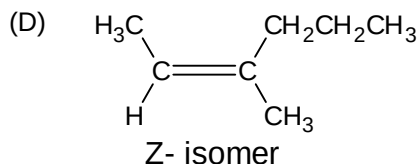
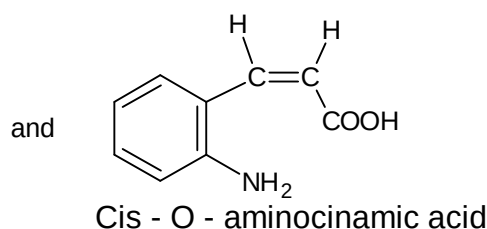
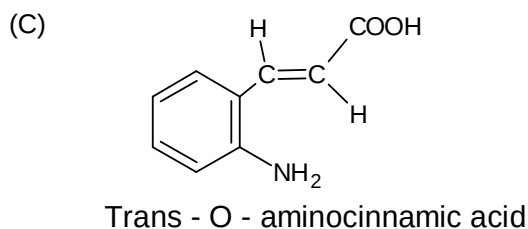
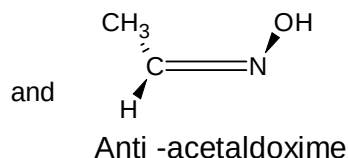
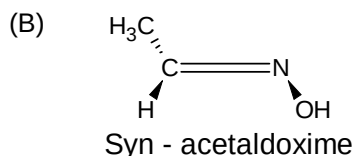
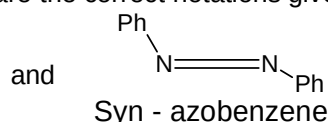
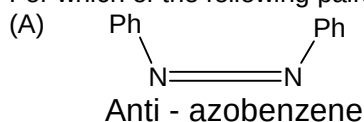
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

24. Which of the following is (are) optically active?



25. For which of the following pairs of compounds are the correct notations given?



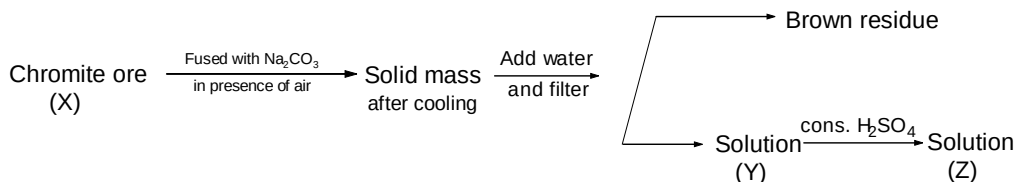
26. Select complex in which no distortion is observed in octahedral geometry.
- (A) $[\text{Cr}(\text{NH}_3)_6]^{3+}$
 (B) $[\text{Co}(\text{CN})_6]^{3-}$
 (C) $[\text{CoF}_6]^{3-}$
 (D) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
27. When red bauxite treated with KOH solution and filtered, then :
- (A) residue contains Fe_2O_3
 (B) residue contains Al_2O_3
 (C) filtrate contains KFeO_2 solution
 (D) filtrate contains KAlO_2 solution
28. Select CORRECT for Hall – Heroult process used for the extraction of Al.
- (A) Aluminium is obtained by reduction of Al^{3+} at cathode
 (B) Oxygen liberated reacts with carbon anodes
 (C) Liquid aluminium is obtained at the top of vessel cell
 (D) Melting point of electrolytic mixture is higher as compared to Al_2O_3
29. In a constant pressure calorimeter with heat capacity of 453 J K^{-1} , 200 mL of 0.862 M HCl and 200 mL of 0.431 M $\text{Ba}(\text{OH})_2$ are mixed. (density = 1 g mL^{-1}) Initially, both solutions are at 20.48°C . Select correct statement.
- (A) $\Delta_r H^\circ$ is -9.86 kJ
 (B) Final temperature is 25.16°C
 (C) Some HCl is left unreacted
 (D) Some $\text{Ba}(\text{OH})_2$ is left unreacted

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

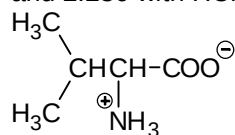
30. Total number of reagents given below reacted with KI (aq.) to produce I_2 , is
- | | | |
|-------------------------------------|-------------------------------------|--|
| CuSO_4 | $\text{Hg}(\text{NO}_3)_2$ | Conc. H_2SO_4 |
| Concentrate H_3PO_4 | $\text{KMnO}_4 / \text{H}^+$ | $\text{K}_2\text{Cr}_2\text{O}_7 / \text{H}^+$ |
| Cl_2 water | Acidify bleaching powder | |
| $\text{Pb}(\text{NO}_3)_2$ | $\text{NaNO}_2 + \text{HCl}$ (dil.) | |

31. Following steps are involved in manufacturing of potassium dichromate:



What is difference in the O.N. of Cr between X and compound of chromium in solution Z?

32. The pH at the half – equivalence point of titration of the Zwitter ion (A), valine 9.714 with NaOH and 2.286 with HCl. Thus, its isoelectric pH is



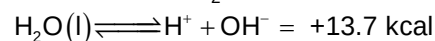
Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 33 and 34

Question Stem

Dissociation of H_2O in H^+ and OH^- is endothermic at 298 K

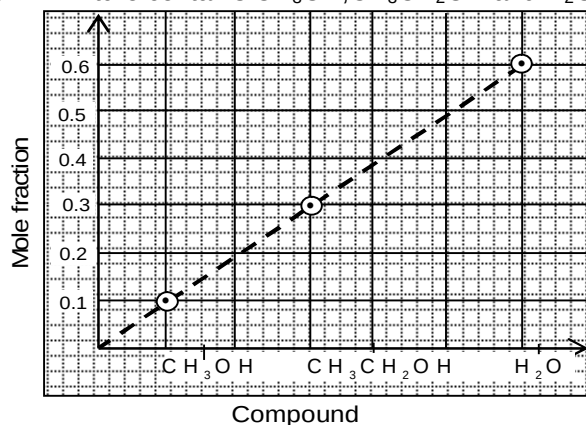


33. ΔG° associated with the equilibrium is _____ kcal.
34. Entropy change associated with formation of H_2O from H^+ and OH^- ions is _____ $\text{cal mol}^{-1}\text{K}^{-1}$.

Question Stem for Question Nos. 35 and 36

Question Stem

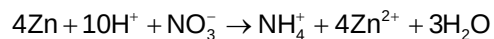
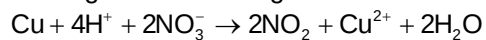
Directions: A mixture contains CH_3OH , $\text{CH}_3\text{CH}_2\text{OH}$ and H_2O and mole fractions are shown below:



35. Molality of CH_3OH in the above case is
36. Mass percentage of $\text{CH}_3\text{CH}_2\text{OH}$ is

Question Stem for Question Nos. 37 and 38**Question Stem**

Directions (Q. Nos. 37 and 38) Brass is a mixture of 90% Cu (63.5) and 10% Zn (65.0). HNO_3 reacts with brass according to the following reactions:



37. What volume (in mL) of 3.0 M HNO_3 can react with 15.0 g of brass?
38. What volume (in L) of NO_2 gas at N.T.P. would be produced?

Mathematics

PART – III

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

39. If $\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$ and $S_1 = \sum_{k=1}^{\infty} \frac{i}{(36k^2 - 1)^i}$, then $S_1 + S_2$ is equal to
- (A) $\frac{\pi^2}{9} - \frac{1}{2}$
- (B) $\frac{\pi^2}{12} + \frac{1}{2}$
- (C) $\frac{\pi^2}{15} + \frac{1}{2}$
- (D) $\frac{\pi^2}{18} - \frac{1}{2}$
40. In a $\triangle ABC$ if $\angle B + \angle C = \frac{2\pi}{3}$ and BI and CI be the angle bisector of angle B and C respectively then the value of $\frac{AP}{BI} + \frac{AQ}{CI}$, where P and Q are feet of perpendiculars from A on BI and CI respectively is
- (A) $\frac{1}{\sqrt{3}}$
- (B) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$
- (C) $\frac{\sqrt{3}+1}{\sqrt{3}-1}$
- (D) $\sqrt{3}$
41. If $[a]$ denotes the integral part of a and $x = a_3y + a_2z$, $y = a_1z + a_3x$ and $z = a_2x + a_1y$, where x, y, z are not all zero. If $a_1 = m - [m]$, m being a non-integral constant, then $a_1a_2a_3$ is
- (A) >1
- (B) >-1
- (C) <1
- (D) <-1
42. Let p be the sum of all possible determinants of order 2 having 0, 1, 2 and 3 as their four elements. Then, the common root ' α ' of the equations $x^2 + ax + [m+1] = 0$, $x^2 + bx + [m+4] = 0$, $x^2 - cx + [m+15] = 0$, such that $\alpha > p$, where $a + b + c = 0$ and $m = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} \frac{r}{\sqrt{n^2 + r^2}}$ (where $[\cdot]$ denotes the greatest integer function)
- (A) ± 3
- (B) 3
- (C) -3
- (D) None of these

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

43. If A, B and C are angles of any one of the triangle such that $\tan A + \tan B + \tan C = 100$. Then,
 (A) there exist exactly three non-similar isosceles triangles
 (B) there exist exactly two non-similar isosceles triangles
 (C) there exist exactly three similar isosceles triangles
 (D) None of the above
44. The equation $(ay - bx)^2 + 4xy = 0$ has rational solutions x, y for
 (A) $a = \frac{1}{2}, b = 2$
 (B) $a = 4, b = \frac{1}{8}$
 (C) $a = 1, b = \frac{3}{4}$
 (D) $a = 2, b = 1$
45. Common tangents are drawn to two circles $x^2 + y^2 = 1$ and $(x - 4)^2 + (y - 4)^2 = 4$. If the intersection points of transverse common tangents with the direct common tangents are A, B, C and D, then
 (A) ABCD is a cyclic quadrilateral
 (B) equation of circumcircle of $\triangle ABC$ is $x^2 + y^2 - 4x - 4y = 0$
 (C) equation of circumcircle of $\triangle BCD$ is $x^2 + y^2 - 5x - 5y = 0$
 (D) ABCD is a square
46. If a, b, c $\in \mathbb{R}^+$ and $a + b + c = 8$ then
 (A) $\frac{a^2 + b^2}{a + b} + \frac{b^2 + c^2}{b + c} + \frac{c^2 + a^2}{c + a} \geq 8$
 (B) $\frac{a^2 + b^2}{a + b} + \frac{b^2 + c^2}{b + c} + \frac{c^2 + a^2}{c + a} \geq 2$
 (C) $a^2b^2 + b^2c^2 + c^2a^2 \geq 8abc$
 (D) $a^2 + b^2 + c^2 \geq 18$
47. If $\alpha, \beta, \gamma, \delta, \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ such that $\sin\alpha + \sin\beta + \sin\gamma + \sin\delta = 1$ and $\cos 2\alpha + \cos 2\beta + \cos 2\gamma + \cos 2\delta \geq 10/3$ then $\alpha, \beta, \gamma, \delta$ must lie in
 (A) $\left[0, \frac{\pi}{12}\right]$
 (B) $\left[-\frac{\pi}{3}, \frac{\pi}{3}\right]$
 (C) $\left[0, \frac{\pi}{6}\right]$
 (D) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

48. Let $f(x)$ and $g(x)$ are two non periodic functions then which of the following may be periodic functions
- (A) $f(x).g(x)$
 - (B) $f(x) + g(x)$
 - (C) $f(x) - g(x)$
 - (D) $f(g(x))$

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

49. P is any point and O being the origin. On the circle with OP as diameter two points Q and R are on same side of OP such that $\angle POQ = \angle QOR = \theta$. Let P, Q, R be z_1, z_2, z_3 such that $2\sqrt{3}z_2^2 = (2 + \sqrt{3})z_1z_3$. The degree measure of θ is α° , then $\frac{\alpha}{3}$ is _____ (in degrees)
50. If the ellipse $x^2 + 81y^2 - 81 = 0$ and the circle $x^2 + y^2 - 9 = 0$ intersect an angle θ in first quadrant, then the value of $(-3 \tan \theta)$ is _____
51. A, B, C and D cut a pack of 52 cards successively in the order given. If the person who cuts a spade first receives Rs. 350 and if the probability of A is λ , then $\left[\frac{350\lambda}{64} \right]$ is equal to (where $[.]$ denotes greatest integer function) _____.

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 52 and 53

Question Stem

$$f(x) = \begin{cases} \frac{2}{\pi} \sin^{-1} \sin\left(\frac{\pi x}{2}\right) & 0 < x \leq 1 \\ (a-1)^2 + (x-1)^2 & x > 1 \end{cases}$$

52. If $x = 1$ is point of local maximum of $f(x)$, then sum of all integral value of $|a|$ is _____
53. Area bounded by curve $y = f(x)$, line $y = 1$ and x-axis is _____

Question Stem for Question Nos. 54 and 55

Question Stem

Consider a curve C passing through the point (1, 1) and has the property that the perpendicular distance of the normal at any point P on the curve from origin is equal to the perpendicular distance of P from x-axis.

On the basis of above information, answer the following questions :

54. Area of the smaller region bounded by curve 'C' and $x^2 - 3y^2 - 2x + 1 = 0$ is

55. Maximum distance of curve 'C' from any point in set $A = \{(x, y) : y^2 + 4x - 4 = 0; x \in [0, 1]\}$

Question Stem for Question Nos. 56 and 57

Question Stem

The parameters representing the points A, B, C and D of the parabola $y^2 = 4ax$ are given by t_1, t_2, t_3 and t_4 respectively. The chord AC subtends an angle of 90° at points B and D.

56. Value of $\left| \frac{t_2 + t_4}{t_1 + t_3} \right| = \underline{\hspace{2cm}}$
57. Minimum value of $|t_1 - t_3| = \underline{\hspace{2cm}}$